

Allowing for crystalline structure effects in Geant4



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ABSTRACT

In recent years, the Geant4 toolkit for the Monte Carlo simulation of radiation with matter has seen large growth in its diverse user community. A fundamental aspect of a successful physics experiment is the availability of a reliable and precise simulation code. Geant4 currently does not allow for the simulation of particle interactions with anything other than amorphous matter. To overcome this limitation, the GECO (GEant4 Crystal Objects) project developed a general framework for managing solid-state structures in the Geant4 kernel and validate it against experimental data. Accounting for detailed geometrical structures allows, for example, simulation of diffraction from crystal planes or the channeling of charged particle.

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1. Introduction

Physics experiments usually take advantage of the most recent technologies and materials for design and construction. As semiconductor companies invested many resources in R&D in order to produce high quality crystalline materials for information technology, samples of such materials can be easily bought on the market. The widespread use of semiconductors and all the possible forms they can take in an experimental apparatus have brought physicists to adopt such materials for a wide range of applications, from detection to manipulation of charged and neutral beams.

Semiconductor detectors are currently adopted in a wide area of physics applications from track reconstruction in high-energy accelerators (ATLAS [1], CMS [2]) and in detection of astrophysics radiation from the Universe (AMS [3], FERMI-LAT [4]) to dark-matter detection in underground experiments (CDMS [5], COSENT [6]). Crystalline targets are currently adopted as beam optics for high-energy accelerators. Bent crystals have been shown to work as beam manipulator for positive and negative particles [7,8], and as leading actors of an innovative collimation scheme for the LHC synchrotron with the UA9 experiment [9].

A fundamental aspect of the design of an experimental apparatus and the interpretation of results for a successful physics experiment is the availability of a reliable and precise simulation code.

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Monte Carlo simulations of the interaction of particles with matter are usually done with software packages such as Geant4 [10–12] and Fluka [13]. Such Monte Carlo codes are continuously expanded and improved thanks to the collaborative effort of scientists from around the world. Geant4 is a general-purpose object-oriented toolkit, which was initially developed for high-energy physics.

Geant4 has seen a large expansion of its user community in recent years. Applications simulated by Geant4 range from particle transportation in the LHC detectors [1,2], to calculations of dose distribution curves for a typical proton therapy beam line [14], and from radiation analysis for space instruments [15] to early biological damage induced by ionizing radiation at the DNA scale [16]. The signal response of a silicon detector, the diffraction from crystal planes or the manipulation of charged particles by channeling are not available in the current Geant4 package. Indeed, although Geant4 deals with the interaction of particles with matter, the toolkit does not allow the simulation of particles interacting with other than amorphous states of matter.

With software infrastructure inside the Geant4 kernel capable of managing crystal structures, it is possible to develop specific physics models to describe their effects, sometimes using already existing physics processes. The GECO project of INFN is devoted to the development of a general framework for the management of solid-state structures in the Geant4 kernel and to validate it against experimental data. In this paper we present the framework integrated in Geant4, in order to accommodate specific solid-state physics processes.

2. State of the art

The Geant4 toolkit provides a diverse, wide-ranging, yet cohesive set of software components, which can be employed from simple one-off studies of basic phenomena and geometries to full-scale detector simulations for experiments at the Large Hadron Collider and other facilities.

All aspects of the simulation process have been included in the definition and implementation of the software components: the geometry of the system, the materials involved, the fundamental particles of interest, the generation of primary particles of events, the tracking of particles through materials and external electromagnetic fields, the physics processes governing particle interactions, the response of sensitive detector components, the generation of event data, the storage of events and tracks.

The materials are made of a single element or a mixture of elements; elements are made of a single isotope or a mixture of isotopes. Because the physical properties of materials can be described in a generic way by quantities which can be either given directly, like density, or derived from the elemental composition, only concrete classes are necessary in this category. The materials category also implements facilities to describe surface properties for the tracking of optical photons.

Currently, the Geant4 kernel does not handle crystalline structures. The state of matter description of the materials is limited to a set of three values, i.e., solid, liquid or gas, which define the limits for the density in that state.

Use of the singleton design pattern for the management of crystalline structures was first introduced with version Geant4.9.6 in an example application for the description of phonon propagation in a germanium crystal [17]. The description of the crystal was limited to the management of a pre-computed map of group velocities and the phonons were treated as particles. From Geant4.10.1, a model suitable for Geant4 implementation was developed to simulate orientational effects of charged particles interacting with bent crystals under planar channeling and volume reflection [18]. Apart from the official Geant4 release a code for the simulation of neutron diffraction was developed [19].

3. Implementation

The development of the code for the management of crystal structures in Geant4 has strict requirements other than the correctness of the crystallographic description of the material. Indeed, from a semantic and logical computational point of view, a new package for the Geant4 kernel has to be harmoniously

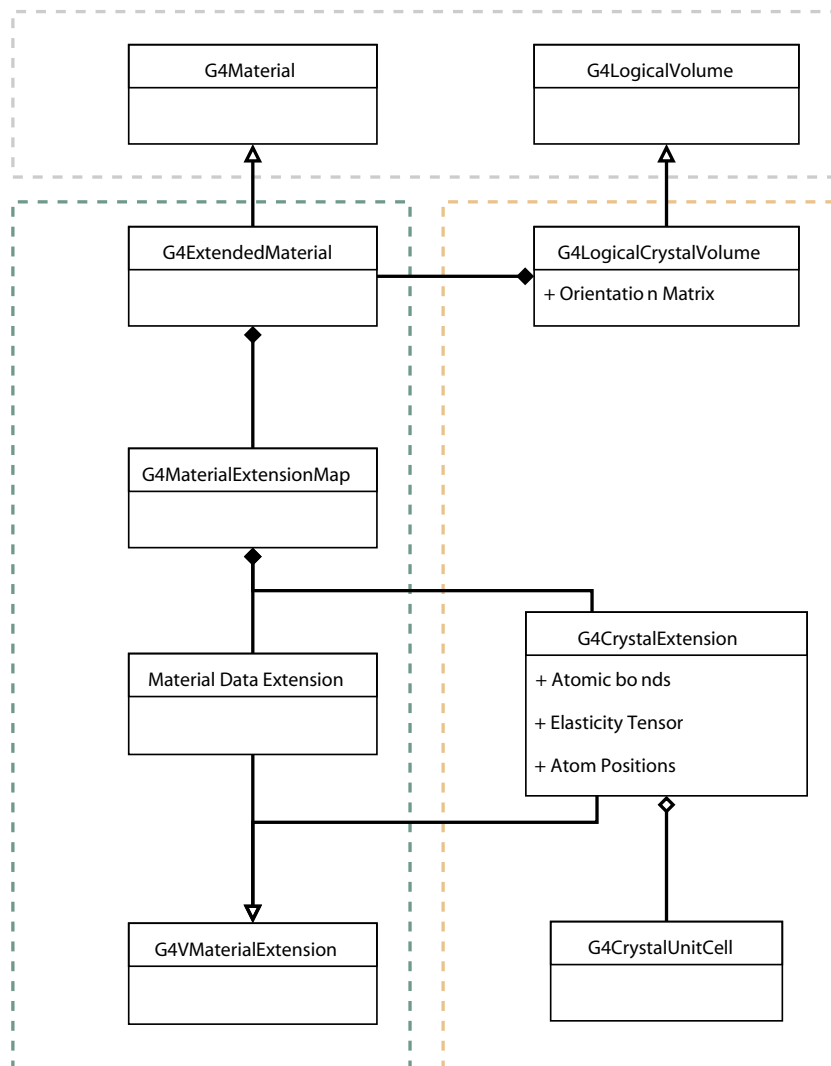


Fig. 1. Class diagram of the implementation of crystal structures into Geant4 with the default Geant4 classes *G4Material* and *G4LogicalVolume* (Default box), the extended classes (Extended box) and the crystal class (Crystal).

integrated into the existing framework in order to be compatible with the remaining code. In addition, because of the open source nature of Geant4 and the ongoing research for a collective development by the international collaboration, the design and the syntax have to be readable by physicists and modifiable by other users and developers of the toolkit. Moreover, the implementation should not negatively impact the computational performance of Geant4, because many users would prefer to use it only optionally.

To fulfil these requirements, the code was designed to be used on top of the existing Geant4 code, with minor modification to the base classes.

The class diagram is shown in Fig. 1. The implementation of solid-state processes in Geant4 requires the addition of two important features, the crystal unit cell with all its parameters and the support for other data required by the processes, i.e. the map of the interplanar electric field in a crystal for the channeling process. Moreover, more than one process may need to add data to a material, as well as processes other than the solid-state ones.

The extended data for a material is stored in a class derived from the virtual class `G4VMaterialExtension`. The `G4VMaterialExtension` virtual class is basically an empty container. The constructor takes a string that is used as the identification name. The string is unique and identifies the extension to the material and can be accessed publicly.

The pointers to the data are stored in a class derived from the `G4Material` class, namely `G4ExtendedMaterial`. The `G4ExtendedMaterial` class collects a `std::unordered_map` of pointers to concrete instances of `G4VMaterialExtension` with a string as the key. The usage of the `std::unordered_map` allows for reducing the search time, since the key is a hash.

The `G4CrystalExtension` class is a derived class of `G4VMaterialExtension`. All the public methods and data members of the `G4Material` class are accessible via a pointer to the related `G4Material`. The `G4CrystalExtension` class collects information on the physics properties of a perfect crystal. In particular, the class contains a pointer to a `G4CrystalUnitCell` object, the elasticity tensor, a map of `G4CrystalAtomBase` objects associated with a `G4Element` and a vector of `G4AtomicBond`.

The `G4CrystalUnitCell` class collects information on the mathematical description of the crystal unit cell, i.e. the sizes and the angles of the unit cell, the space group, the Bravais lattice and the lattice system. The class also provides methods for the calculation of the volume in the direct and reciprocal space, the spacing between two planes, the angle between two planes, and for the filling of the reduced elasticity tensor. Many functions were originally developed for the G4CMP package [17].

The `G4CrystalAtomBase` class stores the position of atoms in the crystal unit cell. Since the `G4CrystalAtomBase` class is

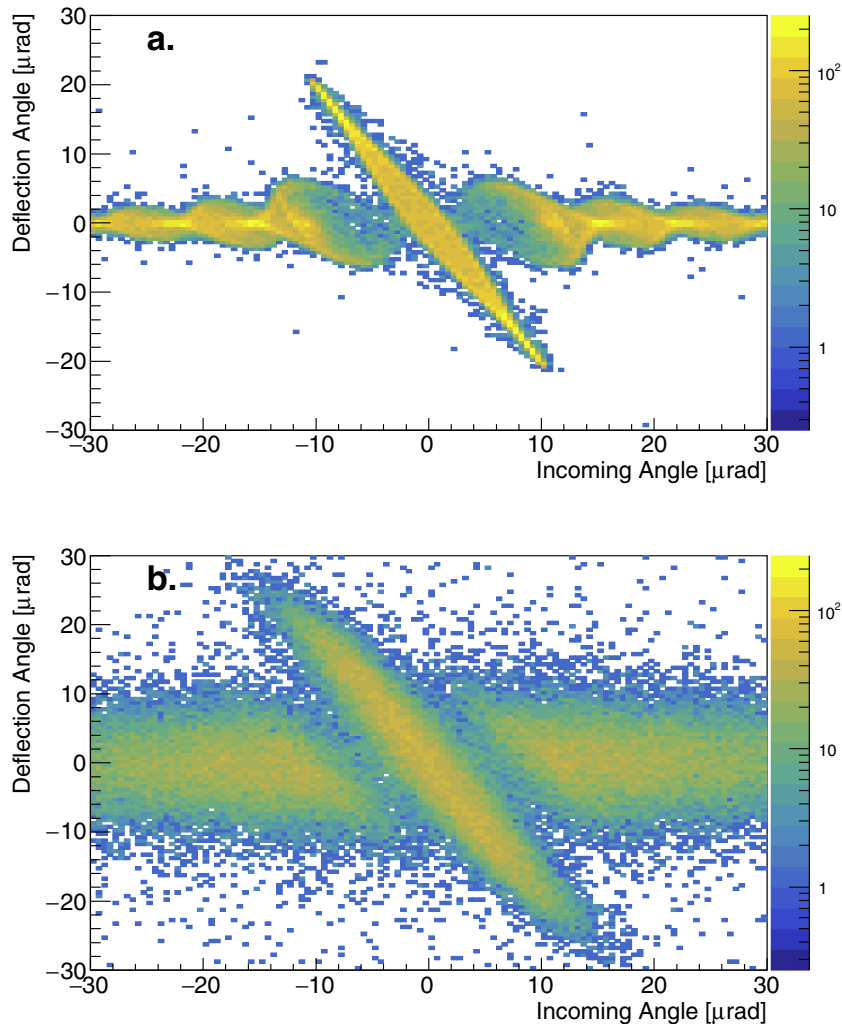


Fig. 2. Geant4 simulation of the distribution of the particle deflection angle as a function of the incoming angle is shown for 400 GeV/c proton beam interacting with a 26.5 μm Silicon crystal. The simulation was carried out with the contribution of the crystal only (a.) and with the contribution of the whole experimental setup (b.) and is in good agreement with experimental data and Monte Carlo simulations in literature [20]. The counts on the Z-axis are in arbitrary units.

mapped to a `G4Element` in the `G4CrystalMaterial`, each `G4Element` should have an associated `G4CrystalAtomBase`.

The `G4AtomicBond` class contains information on the atomic bond in the crystal. For each instance of the class, two `G4Elements` have to be specified as well as the atom number in the `G4CrystalAtomBase` associated to the `G4Element`.

The `G4CrystalExtension` constructor takes as argument a pointer to a `G4Material` object and has to be registered to the `G4ExtendedMaterial` to which it is attached. Then, a `G4CrystalUnitCell` can be added to the `G4CrystalExtension` as well as the extensions needed by the processes.

The implementation of the crystal structure also requires the definition of the orientation of the crystalline structure with respect to the solid to which it is attached. Such information is stored in a derived class of a `G4LogicalVolume`, the `G4LogicalCrystalVolume`. In fact, the `G4LogicalCrystalVolume` accepts only a pointer to a `G4ExtendedMaterial` in its constructor. By convention, the crystal $\langle 100 \rangle$ direction is by default set parallel to the $[1, 0, 0]$ direction in the Geant4 reference system, and the $\langle 010 \rangle$ axis lays on the plane which contains the $[1, 0, 0]$ and $[0, 1, 0]$ directions in the Geant4 reference system.

As an example of the usage of the framework, the code for the channeling process at high-energy [18] has been modified to make use of the new `G4CrystalExtension` class. The `G4ChannelingMaterialData` derived from the `G4VMaterialExtension` class collects info on the crystal electric field, potential, nuclei and electron density under the continuum potential approximation [21]. Such data were calculated with external software [22] and loaded into the `G4ChannelingMaterialData` object during the material construction. The algorithm for the computation of the particle trajectory in the crystal field was updated [23] in Geant4.10.2 with respect to the first release in Geant4.10.1. The integrated framework developed for the crystalline structures was used together with the `QGSP_BIC` physics list and the `G4EmStandardPhysicsSS` electromagnetic physics constructor. The usage of a single scattering electromagnetic model is mandatory since the multiple scattering models implemented in Geant4 modify not only the particle momentum, but also its position, causing the channeling process to track the particle trajectory improperly. As a result, the simulation of the channeling effect may take advantage of the physics processes already implemented in the Geant4 toolkit, but also of all other Geant4 utilities, as the interface to biasing techniques [12]. Indeed, biasing technique is used to scale the cross section of existing Geant4 processes proportionally to the average density of nuclei and electrons experienced by a channelled particle and to preserve the code re-usability for future releases of Geant4. As an example of calculation, in a recent paper, the code was used to simulate a 2.01 mm long bent crystal with 120 GeV/c positrons and electrons [24]. Fig. 2 shows results from a simulation of a 400 GeV/c proton beam interacting with a 26.5 μm Silicon crystal. In the figure, the distribution of the particle deflection angle as a function of the incoming angle is shown. The simulation is in good agreement with experimental data and Monte Carlo simulation in literature [20].

4. Conclusions

The Geant4 code has been upgraded to support the description and simulation of particle interactions with crystalline structures. This work was done as part of the GECO project. The implementation of these features required minor changes to the existing Geant4 code. The development makes use of derived class of the `G4Material` and `G4LogicalVolume` base classes. The informa-

tion related to the crystal unit cell is stored in the `G4CrystalExtension` class, while the orientation of the crystal with respect to the `G4VSolid` to which it is attached are in the `G4LogicalCrystalVolume`. Since each process may need additional information, the `G4ExtendedMaterial` class allows also for the storage of additional material data that can not be classified as crystal data. The class can also be used by other developers, since it depends only on the `G4Material` class. Geant4 10.3 will include support for crystal structures as reported in this paper, allowing for the implementation of processes related to solid-state physics in existing and future simulations.

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